

Reaction Wheel Array: Cone Configuration

DEIMoS ADCS

1 Geometry

The array uses three reaction wheels mounted on a cone. Each spin axis is canted by

$$\theta = \arccos\left(\frac{1}{\sqrt{3}}\right) \approx 54.7356^\circ$$

from the body $+z$ axis, with the three axes spaced 120° apart in azimuth. Wheel 1's in-plane projection is aligned with body $+y$. The unit spin axis of wheel i is

$$e_i = \begin{bmatrix} \sin \theta \sin \phi_i \\ \sin \theta \cos \phi_i \\ \cos \theta \end{bmatrix}, \quad \phi_i \in \{0^\circ, 120^\circ, 240^\circ\}.$$

The cant angle θ is not a free design parameter: it is the unique angle for which three axes spaced 120° apart in azimuth are mutually orthogonal, i.e.

$$W^T W = I, \quad W = [e_1 \ e_2 \ e_3].$$

This is verified numerically at construction time (Gram matrix check).

2 Wheel Momentum

Stacking the axes into the 3×3 matrix W , the wheel angular momentum expressed in the body frame is

$$h_w = W (I_w \odot \Omega) = \sum_{i=1}^3 I_{w,i} \Omega_i e_i,$$

where $I_{w,i}$ is the i -th wheel inertia and Ω_i its spin rate relative to the body.

3 Control Allocation

Given a desired body control torque τ_c , conservation of angular momentum requires

$$Wg = -\tau_c \implies g = -W^+ \tau_c,$$

where g is the vector of motor torques and W^+ is the Moore–Penrose pseudo-inverse. Since the cone axes are orthonormal, W is square and orthogonal, so

$$W^+ = W^T.$$

The torque actually delivered to the body is $\tau_{\text{body}} = -Wg$, equal to τ_c whenever no wheel torque is clipped.

4 Wheel Dynamics

Neglecting wheel back-coupling ($I_w \ll J$),

$$\dot{\Omega}_i = \frac{g_i}{I_{w,i}}, \quad |g_i| \leq \tau_{\max}, \quad |\Omega_i| \leq \Omega_{\max}.$$